NBA Shot Quality and Probabilities

Abstract

This project delivers a model that predicts NBA shot make probabilities from derived player tracking data. Using the model, we are able to evaluate each shot taken, deriving a shot quality evaluation for each player. Behind the model, a mix of bayesian learning and contextual metrics trained on player tracking data assigns a probability to each anonymized player shot, giving each shot a simulated league average probability.

1. Introduction

The goal for this project was to predict shot outcomes given contextual tracking data surrounding each shot. The contextual data included player location, defender distance, player speeds, shot types, and more. Throughout this paper, I will explain the features, methodology, model ramifications, and analyze the model on our data. This paper contains only my own work with references at the bottom, and is formatted to be completely reproducible.

2. Analysis

2.1 Exploratory Contextual Feature Selection

An easy starting point is to just use every feature in the dataset without context. So I trained a simple regression model on each feature to predict the shot outcome, and each feature had a significant p-value so clearly each feature was significant to explaining the outcome of a shot. The log-loss for the baseline model was 0.64.

The data given was hashed to maintain anonymity, so we had no idea which season, team, or players were involved. But through EDA, I was able to find a gaussian distribution of player shots taken as well as shot efficiency. As well as a significant difference in shot selection for each season. So when splitting our training data, we took ~70% of shooters from each season into our training set, and 15% from each season into each of our validation and test sets.

My initial thoughts were the further a shot is, the less likely it is to make the shot. And the more defenders that are closer to a player, the harder it is to make a shot. We can see below that uncontested shots (blue) have a much higher efficiency than contested shots, especially in the restricted area (first dotted line). Now that levels off around the 3 point mark (second dotted line) which is expected since shot contests in the restricted area are much closer than 3 point contests, and every player is expected to make shots in the restricted area. But not every player can shoot 3's regularly, and 3 point shooters are expected to shoot 3s well contested and uncontested.

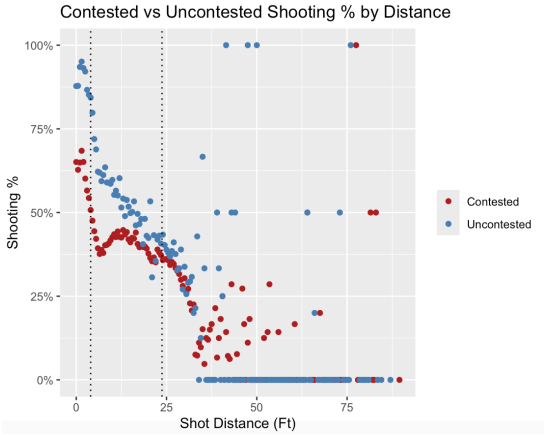


Figure 2.1

With player tracking data, we know it's not perfect and locations can be off. So I utilized a standard shot location grouping method popularized by Kirk Goldsberry, grouping shots into hexagonal regions, then aggregating statistics based on each hexagon. We can see the plots below showing the average FG% and variance of each hexagon.

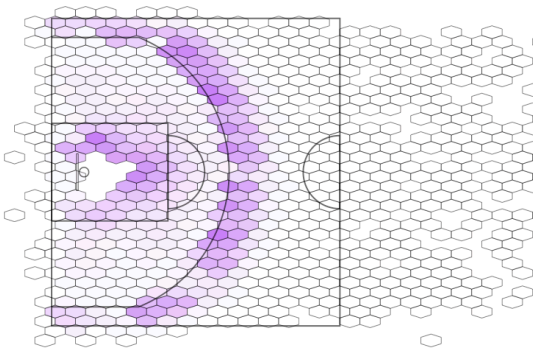


Figure 2.2

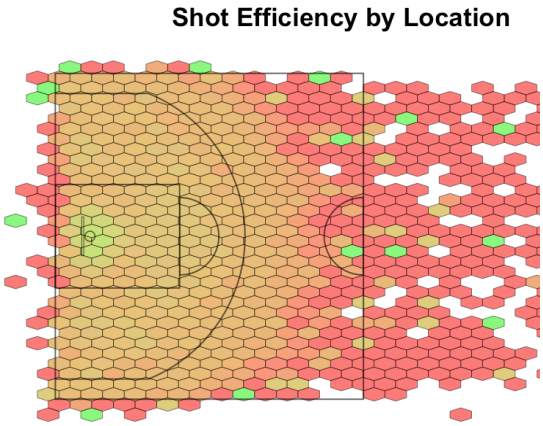
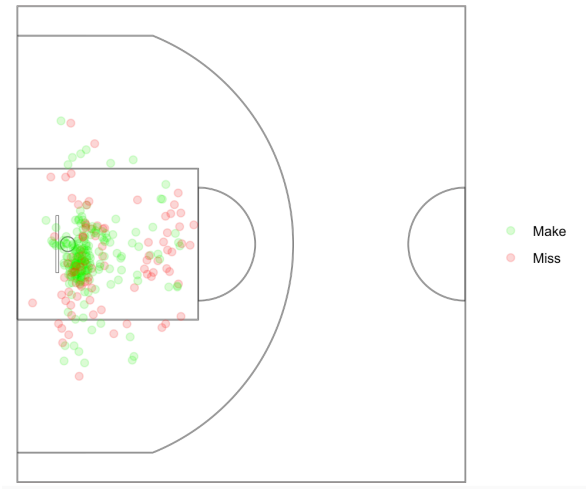


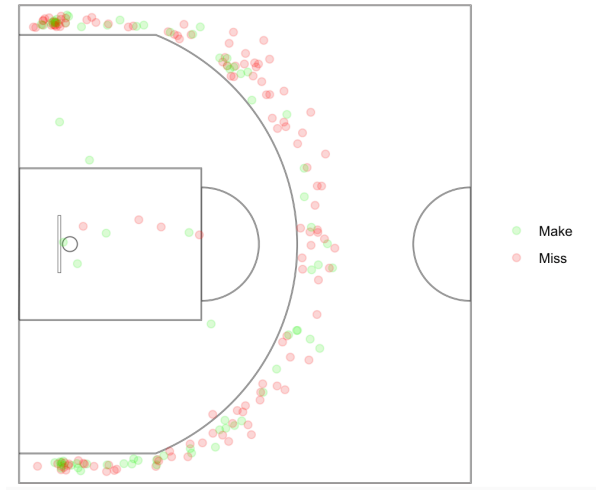
Figure 2.3

2.2 Feature Engineering

Seeing the variance levels in each zone, we can see that players have different skillsets. Steph Curry shooting from 30 feet away has a different probability than Rudy Gobert from the same distance. A center won't shoot as many 3's as a 3&D player. Unfortunately we don't have player height or position in our data. And a 3 point specialist will shoot more from a location he shoots well in, regardless of the defender contest, so player shot volumes will be skewed across the court. You can see below player 'a2db3a89dd' on the left only shot two pointers, while player '1a25ca117f' shot mainly only from 3.



Player *a2db3a89dd* shot chart



Player *1a25ca117f* shot chart

This introduces high variance and noise in our data, where a benchwarmer can come into garbage time, shoot 10 wide open 3's and miss all of them from a spot that league average players convert at a high percentage. Similarly he can bank-in one shot from half court while missing every lay up attempt and we don't want a model thinking he is the greatest 3 point shooter of all time. A player may be forced to shoot a shot he normally doesn't take because of the shot clock expiring.

2.3 Bayes Hierarchical Model

So I decided to implement a Bayesian beta-binomial hierarchical model. Implementing an empirical Bayes model will help estimate a global league prior and shrink each player's make percentage per region toward the league average. We establish 'region' as either (1) restricted area, (2) midrange, or (3) three pointer. The reason we shrink to each region is so that we get the league average of that region, not of the whole dataset. You can see by the plot below:

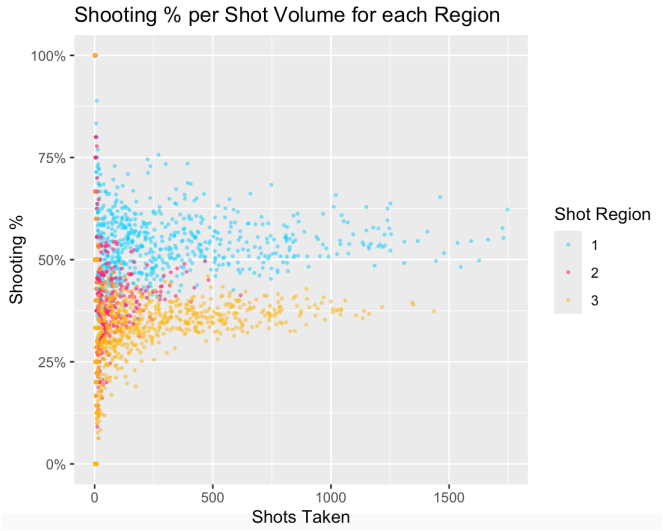


Figure 2.6

The shot location is very important and we cannot shrink a player's restricted area average based on the league 3pt average. To calculate the prior, we get each player's shooting percentage for each zone *i*, and filter to sufficient number of shots (>30 shots) to make sure the player has shot enough shots to have a stable shooting percentage. Then for each zone *i*, we use the beta-binomial method of moments to calculate α_i and β_i and establish the distribution given the mean and variance. Then we implement a jackknife (Leave-One-Out) method to prevent data leakage, calculating the prior using the formula:

i = current zone
j = current shot
k_i = player total makes in zone *i*,
n_i = total shots in zone *i*
$y_{i\,j}$ = the outcome of shot *j* in zone *i*.

$$Shrunk_Skill_{i,j} = \frac{(k_i - y_{ij}) + \alpha_0}{(n_i - 1) + \alpha_0 + \beta_0}$$

When we evaluate our test set we calculate the player's shrunk skill to be the prior league average, thus allowing us to calculate the league average probability for a given shot. You can see the effect where the left plot shows the shot% distribution before we perform the shrinkage for a zone, and then the distribution afterwards.

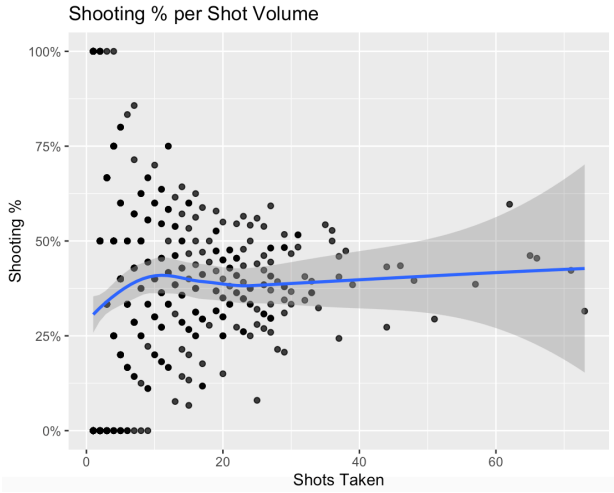


Figure 2.7

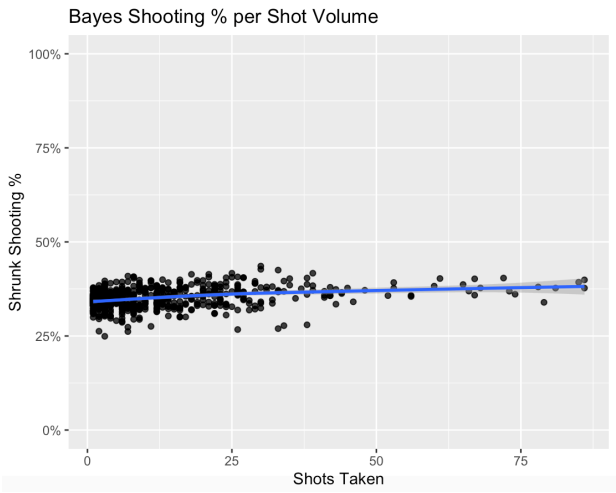


Figure 2.8

The issue that doesn't solve is the model may penalize zones with an imbalance in shot volume distribution. One player could shoot 100 corner 3's at 75% efficiency while 10 others shoot 10 shots at 30%, we don't want the model thinking that the average is 52.5% when the average is truly around 30%. So I decided to introduce sample weights for each shot. We calculate weights for each player as:

w_{pi} = weight for player p, shot zone i

n_{pi} = number of shots taken by player p in zone i

$\alpha = 0.6$

$$w_{p,i} = \frac{n_{p,i}^{1-\alpha}}{n_{p,i} + 6}$$

The weights correct the sampling bias so that each player's contribution to a shooting region is equalized. If one player shoots 100 shots in one particular zone at a high efficiency but the rest shoot minimally at a low efficiency, the model won't penalize the less efficient players just because one player is above average at a high volume. The weights will minimize the effect of the above average player.

This solves our two biggest problems: noise introduced by imbalance in shot volumes, and that some players are more skilled than others. There are other problems like skilled players tend to face tighter defenders but still convert at a high level, but we do not have information

about player and defender characteristics other than distance. Now I wanted to add two more features: hex_number and shot degree angle. Hex number is just the cluster number of the location of the shot, which helps group the location x/y. Shot degree angle is the angle of the shot location to the basket. We found that it is harder to make a shot when you are shooting at a side angle of the basket than if you are shooting straight on. Adding the feature '*angle_deg*' can help the model learn that aspect.

2.3 Model Design

The model I decided to use was a gradient boosted decision tree (GBDT). A GDBT is efficient for large data, can capture complex non-linear boundary effects, and optimizes using log loss via cross entropy.

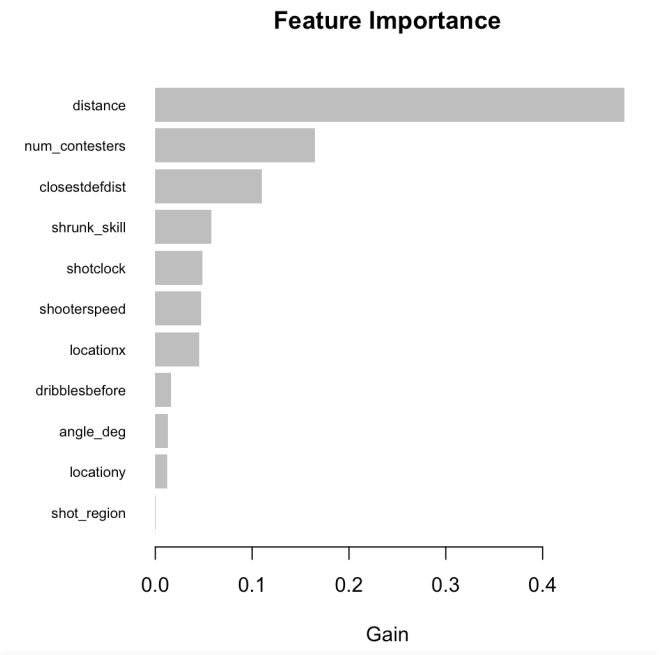
Through EDA we finalized our feature list which included '*num_contesters*', '*contested*', '*gamestate*', '*shottype*', '*distance*', '*locationx*', '*locationy*', '*dribblesbefore*', '*shotclock*', '*closestdefdist*', '*shooterspeed*', '*angle_deg*', '*hex_number*', '*shot_region*', '*shrunk_skill*'. We found those features to be highly significant to explaining a shot outcome probability.

In our model we will implement weights to assist in calculating the error of each row. But we also want to implement monotonic constraints in our model. As a hooper, we can say that generally, the closer a defender is to you, the harder it is to make a shot (while controlling for all other factors). We also can establish that higher skilled players have a higher probability to make a shot when controlling for all factors. And lastly the more people that contest your shot, the harder it is to make the shot. So we put a monotonic constraint +1 on '*closestdistdef*' and '*shrunk_skill*', and a constraint of -1 on '*contested*'.

3. Results

3.1 Model Loss

Our model returns a log loss of 0.631 on our training dataset, with a 0.06 improvement from a naive baseline model. Here is the variable importance plot:



We can see that distance away from the basket is the most important feature, followed by number of contests and the closest defender's distance. Then shruk skill, shot clock, speed, and location x were next with around the same importance level.

3.2 Shot Quality

Since we have the probability of a shot from a league average shooter's perspective, we can establish the probability as the shot quality. A shot with a high quality is a shot that has a high probability. A shot with low quality is a shot with low probability. We can then use the outcome of the shot to evaluate the player's shooting skill (SS). To do this we calculate:

$$SS = outcome - probability$$

We can evaluate players by if they convert a quality shot. If the player misses a high quality shot, that is a low shooting skill. If a player makes a low quality shot, that is a high shooting skill. Obviously there is more to be done here as a player may be lucky/unlucky, but if we calculate the average shot skill over made and missed shots, we can get a general overview of how good a player shoots from a given location.

3.3 Shot Quality & Skill Results

We could look at every region, every defender, every type of shot but that would be overkill. So we focus on 3 pointers, as three pointers have become the modernized shot. Here are the top shooters from 3 with over 30 shots for contested shots (defender < 4ft) based on positive shot skill (convert low probability shots):

season_id	shooter_id	shot_region	defender	shot_skill_pos	shot_skill_neg	shots	pct
fe055	5da155dd38	3	Contested	0.7600714	-0.2528907	35	0.2857143
2676a	e44e631545	3	Contested	0.7328365	-0.2714271	51	0.2352941
fe055	aa40f18608	3	Contested	0.7246408	-0.2588622	56	0.3392857
fe055	d388709b71	3	Contested	0.7238224	-0.2526839	58	0.4827586
fe055	81bc879ea4	3	Contested	0.7222060	-0.2774262	37	0.3243243

As well as the worst shooters with the same parameters based on negative shot skill (miss high probability shots):

season_id	shooter_id	shot_region	defender	shot_skill_pos	shot_skill_neg	shots	pct
fe055	ce6eb0b3c1	3	Contested	0.7069837	-0.3098065	106	0.4150943
fe055	26417591c3	3	Contested	0.6636517	-0.3093076	33	0.2424242
2676a	6a403360ae	3	Contested	0.7105000	-0.3077712	31	0.2580645
fe055	80db449b61	3	Contested	0.6890583	-0.3044121	160	0.3500000
2676a	6f82ac113c	3	Contested	0.6936603	-0.2993261	169	0.3491124

Similarly the best shooters who convert low probability open shots:

season_id	shooter_id	shot_region	defender	shot_skill_pos	shot_skill_neg	shots	pct
fe055	22dc06c5ce	3	Open	0.6783487	-0.3415901	43	0.2790698
fe055	2e36eb87f7	3	Open	0.6629793	-0.3331825	52	0.3076923
fe055	f6dc67d6b8	3	Open	0.6616656	-0.3443797	104	0.2884615
fe055	9573c4add7	3	Open	0.6545987	-0.3300713	367	0.4059946
fe055	a882bb0f01	3	Open	0.6542944	-0.3377886	107	0.3831776

And the worst shooters who miss high probability open shots:

season_id	shooter_id	shot_region	defender	shot_skill_pos	shot_skill_neg	shots	pct
2676a	191bd9e2f3	3	Open	0.5836737	-0.4135357	233	0.3476395
2676a	8f2113635e	3	Open	0.5943882	-0.3964926	155	0.3677419
2676a	14dfd811eb	3	Open	0.6188673	-0.3962365	31	0.3548387
fe055	d804f7f166	3	Open	0.5982042	-0.3954417	99	0.2929293
2676a	202f899c0d	3	Open	0.6011172	-0.3951872	315	0.4190476

Let's look at shooter *d38709b71* who had an average added shot skill% of 0.72. Here is his shot chart and shot skill chart.

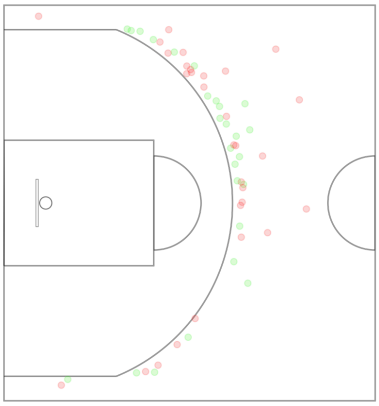


Figure 3.5

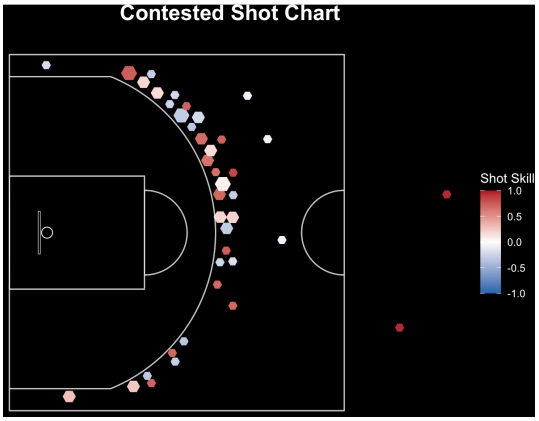
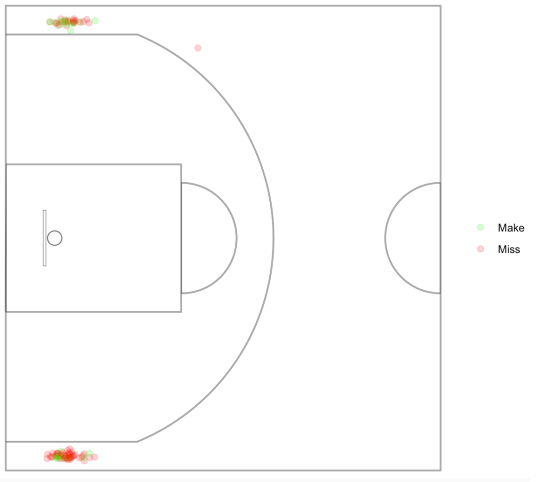


Figure 3.6

We can see that they shot very well contested from 3 (48.3%). If we look at their contested shot skill chart on the right, we can see that they shoot a high amount of contested 3's from the right wing and they convert low probability shots (the larger the hex's the more shots in the area, and the red color means a made low probability shot, blue means missed high probability shot). Obviously this shooter is good at making contested 3's and he knows it and shoots them at a high volume, at a high efficiency and skill rate.

Now let's look at the worst open shooter who doesn't make high probability shots:



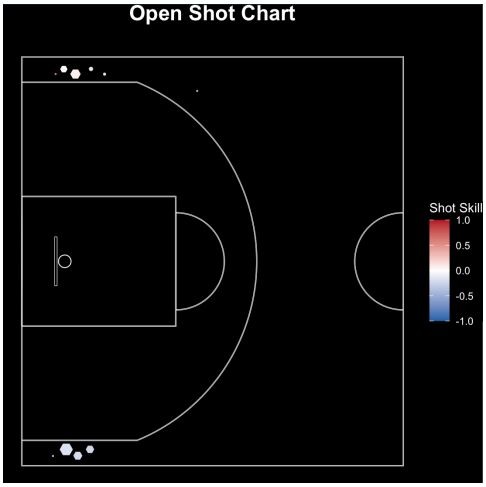


Figure 3.7 Figure 3.8

This shooter is a not so corner 3 specialist, shooting only 29% from 3 and only taking wide open shots from the corner. His left corner is especially bad, missing almost every shot with a high probability.

4. Discussion

4.1 Future Considerations

- If I had more time, I would spend more time researching more features, adding in interaction features such as *is_corner_3* or *dist_to_def_ratio*, and adjusting the model design / hyperparameters.
- I would run K-Fold CV on my training data but wasn't able to due to time constraint and due to the fact I would have to re-model the bayesian shrinkage to each fold.
- Add in player characteristics such as position, defender height, shooter height
- Add in shot events. A shooter shouldn't be penalized for getting blocked against an above average defender.
- Add in the defender angle. This dataset did not provide context on the positioning of the defender such as behind, in front, side. Only the distance was provided.
- Add in shooter handedness. Different handed shooters prefer different locations that tend to have higher shooting percentages.
- Something that I can't think is quite possible to calculate yet is how off balance a shooter is. If a shooter is off balance they should have a lower chance of making the shot, but it is difficult to quantify that.
- Add in player playing time. This can help us generalize their skill.
- I would love to do more analysis on the shot quality and shot skill
- Redefine shot skill. Clearly *outcome - probability* alone does not explain a player's shot. Each shot probability is modeled around the league average so a probability around the league average is a good thing, even if it were 35%.

4.2 Pitfalls

- When binning hex groups for test (submission) dataframe and train datasets since the test dataframe had points that train did not see. So I removed *hex_number* as a feature since it didn't seem to carry a huge importance either.
- It was hard to split the data into their sets since I wasn't sure which season came before the other.
- It was hard to generalize that players take more shots in their 'spots'.
- It was hard to take into account that defenders treat shooters differently based on their role/status/skill. 3 point specialists are usually guarded tighter but they usually shoot at a higher rate anyway.

5. References

Olivier Chabot, Spatial Analysis of Basketball Shots in R, 2021
Kevin Ross, Introduction to Bayesian Hierarchical Models, 2024
SQ, Shot Probability Model Whitepaper

https://github.com/thunder-applications/analyst-intern-2027-changian8-488b3614/tree/cf358ba10eba94b51ffc4dc966f0cefd6ba66503